ORF309 F2025 Midterm1

October 2025

Problem 1. A factory produces widgets on two assembly lines, Line A and Line B.

- Line A produces 60% of the total widgets and has a defect rate of 5%.
- Line B produces the remaining 40% of widgets and has a defect rate of 10%.

All widgets are sent to a central quality control station for testing.

(a) What is the probability that a randomly selected widget is defective?

(b) You are handed a widget that has been identified as defective. What is the probability it came from Line A?

(c) A customer receives a shipment of 10 widgets, selected at random from the entire production. What is the probability that this shipment contains exactly 3 defective widgets?

Solution. Let A denote the event that a randomly selected widget is from Line A, and let D denote the event that a randomly selected widget is defective. Then according to the problem assumptions,

$$\mathbb{P}[A] = 0.6, \quad \mathbb{P}[A^c] = 0.4, \quad \mathbb{P}[D|A] = 0.05, \quad \mathbb{P}[D|A^c] = 0.1.$$

(a) By the law of total probability,

$$\mathbb{P}[D] = \mathbb{P}[A] \cdot \mathbb{P}[D|A] + \mathbb{P}[A^c] \cdot \mathbb{P}[D|A^c] = 0.6 \cdot 0.05 + 0.4 \cdot 0.1 = 0.07.$$

(b) By the definition of conditional probability,

$$\mathbb{P}[A|D] = \frac{\mathbb{P}[A \cap D]}{\mathbb{P}[D]} = \frac{\mathbb{P}[A] \cdot \mathbb{P}[D|A]}{\mathbb{P}[D]} = \frac{0.6 \cdot 0.05}{0.07} = \frac{3}{7}.$$

(c) As a sum of independent Bernoulli random variables, the number N of defective widgets is distributed as Binomial$(10, 0.07)$. Therefore,

$$\mathbb{P}[N = 3] = \binom{10}{3} 0.07^3 0.93^7.$$

$\square$

Problem 2. ChatGPT (and similar large language models) is still imperfect at solving hard math problems. What further complicates matters is that even with the same fixed prompt, its outputs can vary randomly. Suppose that for a given math problem, ChatGPT's responses from repeated trials are independent and uniformly distributed over the set $\{0, 100, 200, 300\}$. Let Z_n denote the average of the first n responses, and F_n denote the fraction of 300's among them.

(a) Compute the expectation of Z_n.

(b) Compute the variance of Z_n.

(c) What can you say about F_n, if n is a very large number? What about the probability of the event $\{Z_n = 300\}$?

Solution. Let $X_1, X_2, ...$ denote the sequence of answers generated. Then they are independent and identically distributed random variables with

$$\begin{aligned}
\mathbb{E}[X_1] &= \frac{1}{4}(0 + 100 + 200 + 300) = 150, \\
\mathrm{Var}[X_1] = \mathbb{E}[X_1^2] - \mathbb{E}[X_1]^2 &= \frac{1}{4}(0^2 + 100^2 + 200^2 + 300^2) - 150^2 = 12500.
\end{aligned}$$

By definition,

$$Z_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad F_n = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i=300\}}.$$

(a) By the linearity of expectation,

$$\mathbb{E}[Z_n] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[X_i] = \mathbb{E}[X_1] = 150.$$

(b) By the scaling property of variance and the mutual independence of $X_1, X_2, ...,$

$$\mathrm{Var}[Z_n] = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}[X_i] = \frac{1}{n}\mathrm{Var}[X_1] = \frac{12500}{n}.$$

(c) By the law of large number,

$$\lim_{n\to\infty} F_n = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i=300\}} = \mathbb{P}[X_1 = 300] = \frac{1}{4}.$$

On the other hand, as Z_n is the average of $X_1, ..., X_n$ and 300 is the highest possible values for the latter,

$$\mathbb{P}[Z_n = 300] = \mathbb{P}[X_1 = 300, X_2 = 300, ..., X_n = 300] = \mathbb{P}[X_1 = 300]^n = \frac{1}{4^n},$$

which goes to 0 as $n \to \infty$. □

Problem 3. Alice and Bob, two arcade enthusiasts, play a claw machine until each wins their first prize. The probability of winning a prize on any single attempt is $p = 0.2$. Each attempt is independent and costs one dollar.

(a) What is the probability that Alice wins her first prize with exactly 3 attempts, given that her first attempt failed?

(b) Given that Bob's first attempt failed, what is his expected total cost to win his first prize?

(c) On average, how much does the more unlucky player (the one who spends more) expect to pay? In other wods, if N_A and N_B denote the total numbers of attempts made by Alice and Bob until their first win, find

$$\mathbb{E}[\max(N_A, N_B)].$$

Solution. Let N_A, N_B denote the number of trials made by Alice and Bob, respectively, until winning their first prize. N_A and N_B are independent random variables distributed as Geometric(p).

(a) Given her first attempt is failed, we can treat the experiments as restarted from the second attempt. Therefore,

$$\mathbb{P}[N_A = 3 \,|\, N_A > 1] = \mathbb{P}[N_A = 2] = p(1-p) = 0.16.$$

(b) With the same reasoning, we see that the conditional distribution of N_B given $N_B > 1$ is the same as the unconditional distribution of $N_B + 1$. Therefore, as the expectation of Geometric(p) equals $\frac{1}{p}$,

$$\mathbb{E}[N_B \,|\, N_B > 1] = \mathbb{E}[N_B + 1] = \frac{1}{p} + 1 = 6.$$

(c) Let

$$N = \max(N_A, N_B)$$

be the maximum amount paid. Then for any $n \geq 1$,

$$\begin{aligned}
\mathbb{P}[N \geq n] &= 1 - \mathbb{P}[N \leq n-1] \\
&= 1 - \mathbb{P}[N_A \leq n-1,\ N_B \leq n-1] \\
&= 1 - \mathbb{P}[N_A \leq n-1] \cdot \mathbb{P}[N_B \leq n-1] \\
&= 1 - (1 - \mathbb{P}[N_A \geq n]) \cdot (1 - \mathbb{P}[N_B \geq n]) \\
&= 1 - (1 - (1-p)^{n-1})^2 \\
&= -(1-p)^{2(n-1)} + 2 \cdot (1-p)^{n-1}.
\end{aligned}$$

As a result,

$$\begin{aligned}
\mathbb{E}[N] &= \sum_{n=1}^{\infty} \mathbb{P}[N \geq n] \\
&= \sum_{n=1}^{\infty} \left(-(1-p)^{2(n-1)} + 2 \cdot (1-p)^{n-1}\right)
\end{aligned}$$

$$= -\frac{1}{1-(1-p)^2} + \frac{2}{1-(1-p)} = -\frac{1}{0.36} + \frac{2}{0.2} = \frac{65}{9} \approx 7.22.$$

$\square$